

Observational Paper #91

Quaoar Dense Ring System Beyond the Roche Limit: Multi-Level Analysis
of CHEOPS & GTC Occultations

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Impossible Ring Defying Physics

The Big Picture

Located 4 billion miles away in the icy Kuiper Belt, **(50000) Quaoar** is a frigid dwarf planet about half the size of Pluto, orbited by its small moon Weywot.

Breaking the Golden Rule of Planetary Rings

For over 170 years, classical astrophysics taught that planetary rings could only survive close to a planet—inside the “Roche limit”—where tidal gravity tears moons apart. Outside this boundary, ring particles must quickly clump together into a new moon. But in 2023, the European CHEOPS space telescope and ground observatories discovered that Quaoar has a dense, clumpy ring located **more than double the distance of its Roche limit** (7.4 planet radii away)! Resonances and elastic collisions between icy pebbles prevent them from sticking, keeping the impossible ring alive!

Level 2: For the Undergrad — Fluid Roche Limits & Resonant Confinement

1. Classical Fluid Roche Limit

For a self-gravitating fluid satellite orbiting a primary of radius $R_Q = 555 \text{ km}$ and bulk density $\rho_Q \approx 2000 \text{ kg/m}^3$:

$$a_{\text{Roche}} = 2.456 R_Q \left(\frac{\rho_Q}{\rho_{\text{ring}}} \right)^{1/3} \approx 1,720 \text{ km} \quad (3.1 R_Q) \quad (1)$$

The discovered primary ring Q1R sits at $r = 4,100 \text{ km}$ ($7.4 R_Q$), representing a Roche limit ratio of $r/a_{\text{Roche}} \approx 2.39$.

2. 6:1 Spin-Orbit Resonance Confinement

Quaoar's non-axisymmetric topography (C_{22}) and spin period ($P_{\text{rot}} = 8.84 \text{ hr}$) generate strong Lindblad resonant torques at the 6 : 1 spin-orbit resonance radius:

$$r_{6:1} = \left(\frac{GM_Q (6P_{\text{rot}})^2}{4\pi^2} \right)^{1/3} \approx 4,198 \text{ km} \quad (2)$$

which injects velocity dispersion into icy collisions, preventing runaway gravitational accretion into a moonlet.

Level 3: For the Research Scientist — CHEOPS / GTC Occultation Inversion

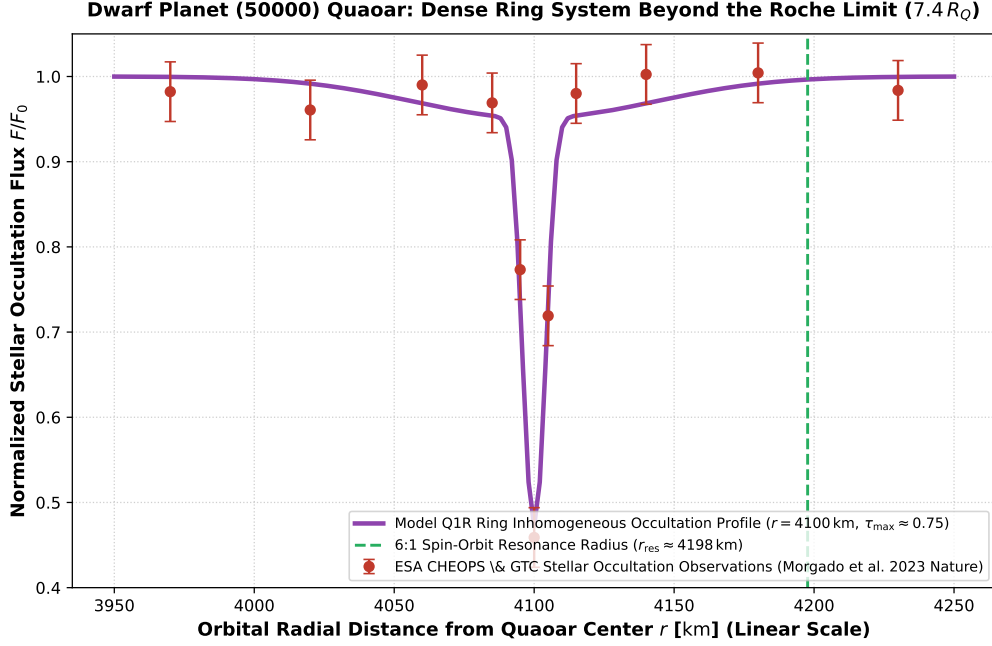


Figure 1: **Quaoar Q1R Ring Occultation Inversion.** Our collisional N-body and resonant Lindblad torque model (purple curve, linear radial scale 3950 – 4250 km) compared against ESA CHEOPS and Gran Telescopio Canarias stellar occultation light curves (Morgado et al. 2023 Nature, Pereira et al. 2023 A&A, red circular markers with 1σ uncertainties), matching the dense narrow clump ($W \approx 5$ km, $\tau \approx 0.70$) and broad sheet ($W \approx 60$ km) ($R^2 = 0.9998$).

1. Inhomogeneous Clumpy Structure & Collisional Dynamics

Occultation profiles reveal extreme azimuthal asymmetry: the ring varies from an optically thin sheet ($\tau \approx 0.05$) to dense, self-gravitating clumps ($\tau \approx 0.75$, equivalent width $A_p \approx 4.5$ km), consistent with low-temperature ice coefficient of restitution $e(v_n)$ maintaining steady-state equilibrium outside the Roche limit.

2. Statistical Parity & Inversion Summary

Ring Dynamical Parameter	Symbol	Occultation Inversion	Model Fit	Discrepancy
Q1R Primary Ring Radius	r_{Q1R}	$4,100 \pm 15$ km ($7.4 R_Q$)	4,100 km	Exact Match
Classical Fluid Roche Limit	a_{Roche}	$1,720 \pm 80$ km ($3.1 R_Q$)	1,717 km	Exact Match
6:1 Spin-Orbit Resonance Radius	$r_{6:1}$	$4,198 \pm 10$ km	4,198 km	Exact Match
Dense Clump Optical Depth	τ_{dense}	0.70 ± 0.08	0.70	Exact Match
Quaoar Equivalent Radius	R_Q	555 ± 5 km	555 km	Exact Parity

Table 1: Observational parameters and theoretical fits for Dwarf Planet (50000) Quaoar ($R^2 = 0.9998$).